Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/27/2006
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Effect of Simulated Microgravity on Gene Expression in the Enteric Pathogen Salmonella typhimurium		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-02
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pierson, Duane (NASA-Johnson Space Center) Ott, MArk (Wyle Laboratories)		
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The effects of spaceflight on the infectious disease process have only been studied at the level of the host immune response, and indicate a blunting of the immune mechanism in humans and animals. Accordingly, it is necessary to assess potential changes in microbial virulence associated with spaceflight which may impact the probability of in-flight infectious disease. We have previously shown that the environment created by simulated or "low shear modeled microgravity" (LSMMG) represents a novel environmental regulatory factor of the virulence of the enteric pathogen, Salmonella typhimurium. The purpose of this proposal is to investigate the ability of LSMMG to affect global gene expression, including regulation of virulence attributes, in S. typhimurium. Specifically, we have used a modified subtraction hybridization approach coupled with whole genome microarray-mediated gene expression profiling to identify and isolate S. typhimurium genetic loci that are uniquely regulated under the conditions of LSMMG, including

Task Description:	those important for the pathogenicity of this organism. We expect that information gained from these studies will prove beneficial in assessing microbiological risks and options for reducing those risks during crew missions. Furthermore, this ground-based study will ultimately provide significant insights into the molecular basis of Salmonella virulence, with potential for vaccine development and other novel strategies for prevention and treatment of disease caused by Salmonella. The development of such countermeasures would ultimately be expected to improve the quality of human life on Earth and during exploration of space. The effects of spaceflight on the infectious disease process have only been studied at the level of the host immune response, and indicate a blunting of the immune mechanism in humans and animals. Accordingly, it is necessary to assess potential changes in microbial virulence associated with spaceflight which may impact the probability of in-flight infectious disease. We have shown that the environment created by modeled microgravity (MMG) represents a novel environmental regulatory factor of the virulence of the enteric pathogen, Salmonella enterica serovar Typhimurium (S. typhimurium). The purpose of this proposal is to identify and characterize S. typhimurium genetic loci that are uniquely regulated in response to MMG, and which may contribute to the pathogenicity of this organism. See Task Description
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The environmental parameters associated with spaceflight represent a unique environment for the microbial pathogen; the presence of which will certainly accompany any human habitat in space. However, the effects of spaceflight on the infectious disease process have only been studied at the level of the host immune response, and indicate a blunting of the immune mechanism in humans and animals. Accordingly, it is necessary to assess potential changes in microbial virulence associated with spaceflight which may impact the probability of in-flight infectious disease. We have provided the first direct evidence of a role for MMG as a novel environmental signal affecting the virulence of the enteric bacterial pathogen, S. typhimurium. Salmonella typhimurium is an important facultative intracellular pathogen that remains a major cause of mortality and morbidity worldwide. In addition, since Salmonella is very closely related to Escherichia coli, the microorganism about which the most is known in terms of physiology, growth, genetics and molecular biology, it can be expected that S. typhimurium will be the most readily and fully understood pathogen in terms of its pathogenic potential in both biochemical and molecular genetic terms. The virulence of bacterial pathogens becomes critically important in individuals who are immunocompromised, as would appear to be the case during space flight. The potential interaction of the crew with pathogenic bacteria in a self-contained environment is increased with the addition of proposed regenerative life support systems, including waste remediation. As the duration of the mission increases, enteric bacteria such as S. typhimurium will inevitably compose a large segment of the bacterial consortia in certain systems. The commencement of long-term missions that will use regenerative systems, such as the International Space Station, creates an urgent need to investigate potential changes in bacterial pathogenicity caused by prolonged conditions of microgravity. This becomes a significant issue to address, especially if host defenses deteriorate during spaceflight. The purpose of this proposal is to elucidate the mechanisms by which altered gravitational forces affect gene regulation and expression of virulence attributes in S. typhimurium. Toward this end, we are using the High Aspect Rotating Vessel (HARV), to simulate conditions of microgravity in order to model the effects of space flight on the molecular mechanisms of bacterial virulence. We are currently using a positive cDNA selection approach (SCOTS) coupled with whole genome microarrayÃ?Â?mediated gene expression profiling to identify and isolate genetic loci in S. typhimurium which are uniquely regulated in response to MMG. We anticipate that this research will ultimately provide significant insights into the molecular basis of Salmonella virulence, and may have potential for vaccine development and other novel countermeasures for prevention and treatment of disease caused by Salmonella.
Task Progress:	We have previously demonstrated that the environment created by low shear modeled microgravity (LSMMG) represents a novel and global environmental regulatory factor of Salmonella virulence (Nickerson, et al., 2000. Microgravity as a novel environmental signal affecting Salmonella enterica Serovar Typhimurium virulence. Infect. Immun. 68:3147-3152). In addition, we have demonstrated that LSMMG-grown Salmonella also displayed increased resistance to environmental stresses (acid, thermal, and osmotic), and global alterations in protein expression levels (Nickerson, et al., 2000. Microgravity as a novel environmental signal affecting Salmonella enterica Serovar Typhimurium virulence. Infect. Immun. 68:3147-3152; Wilson, et al., 2002. Microarray analysis identifies Salmonella genes belonging to the low-shear modeled microgravity regulon. Proc. Natl. Acad. Sci. USA. 99:13807-13812; Wilson, J., et al., 2002. Low shear modeled microgravity alters the Salmonella typhimurium response in an RpoS-independent manner. Appl. Environ. Microbiol. 68:5408-5416). Moreover, Salmonella cultured under LSMMG exhibited increased ability to survive within macrophages (Nickerson, et al., 2000. Microgravity as a novel environmental signal affecting Salmonella enterica Serovar Typhimurium virulence. Infect. Immun. 68:3147-3152; Wilson, J., et al., 2002. Low shear modeled microgravity alters the Salmonella typhimurium response in an RpoS-independent manner. Appl. Environ. Microbiol. 68:5408-5416; Nickerson et al., 2003. Low Shear Modeled Microgravity: A Global Environmental Regulatory Signal Affecting Bacterial Gene Expression, Physiology, and Pathogenesis. (Invited Review) J. Microbiol. Methods. 54:1-11). This global change in phenotype and protein expression by Salmonella in response to LSMMG indicates the existence of a regulatory network we have termed the "LSMMG regulon". Our previous work also indicated that the Salmonella LSMMG response did not require RpoS, the major regulator of environmental signal responses in Salmonella (Wilson, J., et al., 2002. Low shear modeled microgravity alters the Salmonella typhimurium response in an RpoS-independent manner. Appl. Environ. Microbiol. 68:5408-5416). In continuation with the next phase of our objectives for this project, we have used whole genome microarray–mediated gene expression profiling to identify S. typhimurium genetic loci that are uniquely regulated in response to LSMMG, and which may be important for the virulence of this organism. When compared to identical growth conditions under normal gravity (1xg), LSMMG differentially regulated the expression of 163 genes distributed throughout the chromosome, representing functionally diverse groups including transcriptional regulators, virulence factors, LPS biosynthetic enzymes, iron-utilization enzymes and proteins of unknown function. Many of the LSMMG–regulated genes were organized in clusters or operons. The microarray results were further validated by RT-PCR and phenotypic analyses, and they indicate that the ferric uptake regulator (Fur) is involved in the LSMMG response. This work was recently published in the Proceedings of the National Academy of Sciences (Wilson, et al., 2002. Microarray analysis identifies Salmonella genes belonging to the low-shear modeled microgravity regulon. Proc. Natl. Acad. Sci. USA. 99:13807-13812). These studies provide important insight about the Salmonella LSMMG response and could provide

	clues for the functioning of known Salmonella virulence systems, or the identification of uncharacterized bacterial virulence strategies. We are currently in the process of generating mutations in the sequences of those genes identified as uniquely regulated in response to LSMMG, but which encode proteins of unknown function. Following the inactivation of these genes, we will determine their stress resistance profiles and virulence potential. We are also currently studying the expression and regulation of several candidate LSMMG genes identified in Salmonella in vivo and in other organisms as well. Portions of this work have been recently published in the Journal of Biotechnology, BMC Evolutionary Biology, and the Journal of Microbiological Methods (Wilson, J., et al., 2004. VEX-capture: A new technique that allows in-vivo excision, cloning, and broad-host-range transfer of large bacterial genomic DNA segments. J. Microbiol. Methods. 57(3):297-308: Nickerson, C.A., et al., 2004. Microbial responses to microgravity and other low shear environments. Microbiology and Molecular Biology Reviews. 68:345-361; Wilson, J.W., and C. A. Nickerson. 2005. Cloning of a functional Salmonella SPI-I type III secretion system and development of a method to create mutations and epitope fusions in the cloned genes. Journal of Biotechnology. Oct 24; Wilson, J.W., and C. A. Nickerson. 2006. A new experimental approach for studying bacterial genomic island evolution identifies island genes with bacterial host-specific expression patterns. BMC Evolutionary Biology 2006 Jan 5;6(1):2).
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